7.4.15

EE25BTECH11008 - Anirudh M Abhilash

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Question

The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area 154 sq.units. Find the equation of the circle.

Solution

The equations of the diameters are

$$2x - 3y = 5, \tag{1}$$
$$3x - 4y = 7. \tag{2}$$

$$\begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix} \mathbf{c} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \tag{3}$$

Substituting $\mathbf{u} = -\mathbf{c}$:

$$\begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix} \mathbf{u} = -\begin{pmatrix} 5 \\ 7 \end{pmatrix}. \tag{4}$$

Augmented matrix of the system:

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 3 & -4 & -7 \end{array}\right).$$

Step 1:

$$R_2 \to 3R_1 - 2R_2$$

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 0 & -1 & -1 \end{array}\right).$$

Step 2:

$$R_2 \to -R_2$$

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 0 & 1 & 1 \end{array}\right).$$

Step 3:

$$R_1 \to R_1 + 3R_2$$

$$\left(\begin{array}{cc|c} 2 & 0 & -2 \\ 0 & 1 & 1 \end{array}\right).$$

Step 4:

$$R_1 \to \frac{1}{2}R_1$$

$$\left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array}\right).$$

Thus the solution is

$$\mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \tag{5}$$

From the area,

$$\pi r^2 = 154 \quad \Rightarrow \quad r^2 = 49 \quad \Rightarrow \quad r = 7. \tag{6}$$

Compute f (using $f = \|\mathbf{u}\|^2 - r^2$):

$$\|\mathbf{u}\|^2 = (-1)^2 + 1^2 = 2, \tag{7}$$

$$f = 2 - 49 = -47. \tag{8}$$

The general quadratic form is

$$\mathbf{x}^T V\mathbf{x} + 2\mathbf{u}^T\mathbf{x} + f = 0, \tag{9}$$

and for a circle $V = I_{2\times 2}$. Substituting the values:

$$\mathbf{x}^T I\mathbf{x} + 2\begin{pmatrix} -1 & 1 \end{pmatrix}\mathbf{x} - 47 = 0. \tag{10}$$

Expanded,

$$x^2 + y^2 - 2x + 2y = 47, \tag{11}$$

$$\boxed{x^2 + y^2 - 2x + 2y = 47}$$

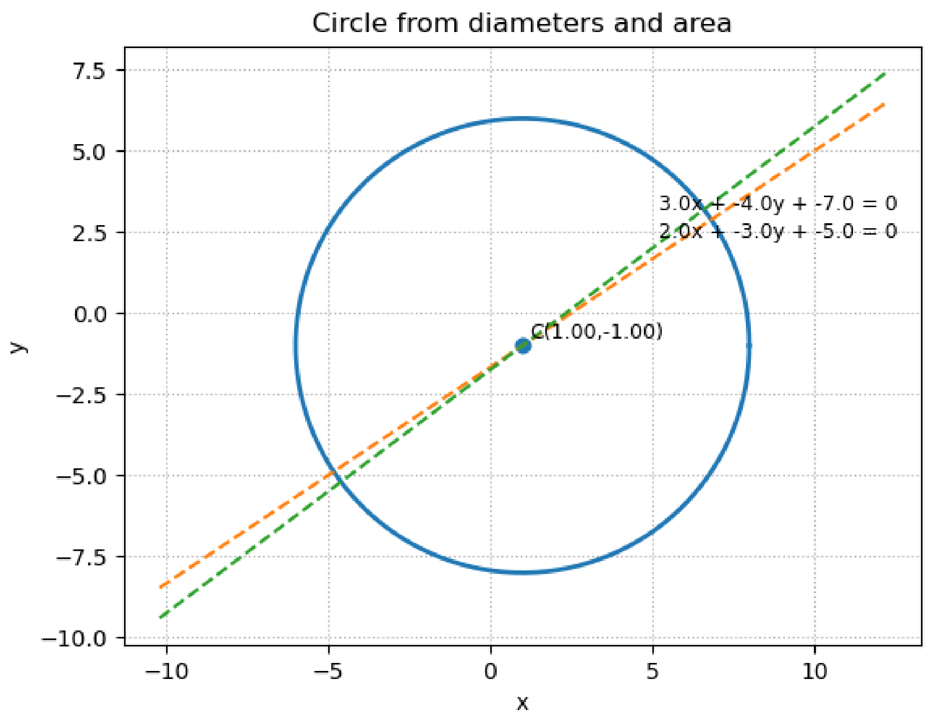

Figure 1: Given Circle